

Creative Retreat — Final Submission

Dubbing Satvic Movement into Regional India, with AI

Role: Founder's Office — AI Specialist

Task: Dub an existing Hindi long-form video into two regional languages, in the creator's own voice, using AI — genuinely watchable and accurate, not just technically possible.

Video dubbed: *Heal Constipation Permanently (5 Cures Nobody Talks About)* — Hindi, 16:12.

1. What was delivered

Deliverable	Where
Telugu dub — Subah's cloned voice, H.264 1080p, music bed intact, 972s	1_Final_Videos/ Satvic_Constipation_Telugu.mp4
Malayalam dub — same spec	1_Final_Videos/ Satvic_Constipation_Malayalam.mp4
YouTube multi-audio tracks (WAV, 48 kHz, -14 LUFS)	1_Final_Videos/YouTube_Audio_Tracks/
Process & reasoning (this doc + the deep writeup)	2_Process_and_Reasoning/PROCESS.md
Pitch deck (technical workflow + business case)	Pitch_Deck.pptx
Native-speaker review kit (460 lines, back-translated, health-claims flagged)	3_Accuracy_Review_Kit/
Reproducible pipeline (scripts, translations, transcript)	5_Pipeline_Workspace/

Cost: ₹0 of the ₹2,500 budget. **Verification:** 0 reference-leaks across all 460 segments.

The final videos are ~339 MB each and are shared alongside this repository (drive / release); 20-second previews with the music bed are in 4_Audio_Samples/.

2. The strategic insight (why this is worth doing at all)

Before choosing any tool, I checked what YouTube already does for free — because a paid pipeline has to justify itself against the platform.

YouTube's free auto-dubbing cannot produce Telugu, Malayalam, or Kannada from a Hindi video. Kannada isn't an auto-dub target from *any* language, and where auto-dub does exist it uses a generic synthetic voice, not the creator's. So the ~190 million people who speak these three languages are invisible to the channel today — purely for lack of an audio track.

Custom, voice-cloned multi-audio tracks are the **only** route to that audience, and almost no Hindi-first channel has taken it. That is the reach opportunity, and it's a durable one.

3. Which tools, and why (the brief's core question)

Step	Tool chosen	Why
Download	yt-dlp	Pulls video + original audio track cleanly. Free.
Transcribe	OpenAI Whisper (medium)	Timestamped Hindi transcript as the translation source of truth. Local, free.
Voice / music separation	Demucs	Splits Subah's voice (clone reference) from the music bed (needed because YouTube's dubbed track replaces the <i>entire</i> mix).
Translation	LLM + a locked glossary + native review	Kept <i>outside</i> the dubbing tool so every word is reviewable before synthesis.
Voice clone + TTS	IndicF5 (AI4Bharat / IIT-Madras)	Open-source, zero-shot voice cloning, native Telugu + Malayalam, runs locally.
Assemble	ffmpeg	Fits each line to timing, remixes the music bed, muxes the video, normalizes loudness.

Why IndicF5 over the alternatives — and the two pivots to get there:

1. **Sarvam AI Dub** won the on-paper comparison (purpose-built for Indic dubbing with voice cloning) — but its dubbing product is **waitlist-gated**, not self-serve. A tool you can't access loses to one you can.

2. **ElevenLabs** has the best clone fidelity and supports both languages, but costs ~₹1,950. I kept it as a **priced fallback**, not the default.
3. **IndicF5** — India's own open-source model — does the exact job (Hindi→Te/MI with voice cloning) for **₹0**, locally. It became Tier 1; ElevenLabs Starter (~₹450) and Creator (~₹1,950) are Tiers 2–3, priced but never needed.

The ₹2,500 budget stopped being a spend and became **an escalation ladder we priced but never had to climb** — the decision to move up it is the native reviewer's, not a default.

Tools evaluated and rejected for this pilot: Rask (no free plan), Camb.ai (self-serve dubbing minutes near-useless), HeyGen/Vozo/Murf (cost or Te/MI-dub coverage), Dubverse (budget math), and the DIY route (Google/Azure/Bhashini TTS) — cheapest per minute but you assemble the whole STT→MT→TTS→sync chain yourself, which is essentially what IndicF5 + this pipeline already is.

4. What didn't work (the honest part)

- **YouTube's caption track** was auto-generated junk and rate-limited; Whisper on the real Hindi audio replaced it.
- **Whisper mangles code-switched English** inside Hindi ("constipation" → कुन्स्टिपेशन) — so transcript repair became step one, before any translation.
- **The first translation was too "pure"** — literary/Sanskritized. Founder feedback corrected it to everyday colloquial (how a Telugu/Malayalam YouTuber actually speaks); all 460 lines were rewritten and the rule locked into the glossary.
- **The reference-voice leak — caught by a native ear, twice.** IndicF5 occasionally echoes its voice-clone reference into a segment's start. My reference clip's tail ("...rectum bent rehta hai", then a "neutral" one whose tail was "...degree angle banta hai") bled into the dub. Worse: because the second leak was partly English, a language-ID scan reported a **false "0 leaks."** A native listener caught what the automation missed. Root cause turned out to be a **badly-cut reference clip**; re-cutting it as a clean, complete sentence dropped the leak rate from **~28% to ~1%** at the source. Both languages were then re-synthesized and verified with a **content-based** detector — final state **0 / 460**.

The lesson threaded through all of these: a detector's green light means nothing until it's stress-tested for false positives *and* false negatives — and for a health channel, the human ear is the ground truth. Which is exactly why the native review is the gate, not the machine.

5. How accuracy was verified

Layered, cheapest-first:

1. **Glossary lock** — brand/ayurvedic terms transliterated, English loanwords kept, all numbers spelled out as words (Indic TTS garbles digits), respectful register.
2. **Source-transcript repair** — every ASR error fixed before translation.
3. **Back-translation on every line** — each of 460 segments carries a literal English back-translation in the review sheet, so meaning is auditable without reading both scripts.
4. **Health-claim flags** — 74 Telugu / 63 Malayalam lines pre-flagged (dosages, durations, do/don't instructions, health promises) for priority review.
5. **Native-speaker watch-down** — script review, then a full watch of the dub. **This is the sign-off.** The review kit ([3_Accuracy_Review_Kit/](#)) makes it a ~40-minute job.
6. **Technical QC** — duration match, sync, -14 LUFS loudness, music bed intact, 0 leaks ([2_Process_and_Reasoning/QC_RESULTS.md](#)).

Please still run the native review before publishing — Satvic's team, or an arranged native speaker per language.

The pipeline gets it fluent and clean; a native ear confirms it's *right*.

6. If we did this at scale (the full library)

- **The pipeline is a queue.** Everything is scripted; human time concentrates only on script-review and the native watch-down — hire one freelance native reviewer per language.
 - **Order by analytics, not alphabet** — dub top watch-time videos first; pick languages from YouTube Geography data.
 - **The glossary compounds** — every reviewed video shrinks the next one's review time. It's the moat.
 - **Keep YouTube's free auto-dub** for the 16 languages it covers from Hindi; spend custom-dub effort only where the platform is blind — Telugu, Malayalam, Kannada, Tamil.
 - **Cost inverts nicely** — one GPU collapses synthesis to minutes; the real cost is native QC (~₹2-3k/video), which is exactly where a wellness brand should spend.
 - **Formalize the voice** — one consented, verified clone per language in a Satvic-controlled account, replacing per-video zero-shot cloning (consistency + clean rights).
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7. Rights & consent

Voice cloning used Subah's own published audio, for Satvic's own content, at Satvic's invitation — and IndicF5's license itself requires permission for any cloned voice, which this use has by construction. For a production rollout: record Subah's formal voice-consent statement and keep the voice assets in a Satvic-controlled account.

8. How to navigate this submission

Start with this document, then: - `Pitch_Deck.pptx` — the presentation (technical workflow + business case). - `2_Process_and_Reasoning/PROCESS.md` — the full detailed writeup, dead ends and all. - `2_Process_and_Reasoning/PRESENTATION_SCRIPT.md` — talking track for the screen recording. - `3_Accuracy_Review_Kit/` — hand to a native speaker to verify accuracy. - `5_Pipeline_Workspace/` — the reproducible pipeline (run order in its README).

Two clean dubs today, in Subah's voice, for ₹0 — and a repeatable system ready for the library.